example, when one electrically charged particle repels another of like electric charge, you can think of it roughly in terms of each particle being surrounded by an electric field—a “cloud” or “mist” of “electric-essence”—and the force each particle feels arises from the repulsion between their respective force fields. The more precise microscopic description of how they repel each other, though, is somewhat different. An electromagnetic field is composed of a swarm of photons; the interaction between two charged particles actually arises from their “shooting” photons back and forth between themselves. In rough analogy to the way in which you can affect a fellow ice-skater’s motion and your own by hurling a barrage of bowling balls at him or her, two electrically charged particles influence each other by exchanging these smallest bundles of light.

An important failing of the ice-skater analogy is that the exchange of bowling balls is always “repulsive”—it always drives the skaters apart. On the contrary, two oppositely charged particles also interact through the exchange of photons, although the resulting electromagnetic force is attractive. It’s as if the photon is not so much the transmitter of the force per se, but rather the transmitter of a message of how the recipient must respond to the force in question. For like-charged particles, the photon carries the message “move apart,” while for oppositely charged particles it carries the message “come together.” For this reason the photon is sometimes referred to as the messenger particle for the electromagnetic force. Similarly, the gluons and weak gauge bosons are the messenger particles for the strong and weak nuclear forces. The strong force, which keeps quarks locked up inside of protons and neutrons, arises from individual quarks exchanging gluons. The gluons, so to speak, provide the “glue” that keeps these subatomic particles stuck together. The weak force, which is responsible for certain kinds of particle transmutations involved in radioactive decay, is mediated by the weak gauge bosons.

Gauge Symmetry

You may have realized that the odd man out in our discussion of the quantum theory of the forces of nature is gravity. Given the successful approach physicists have used with the other three forces, you might suggest that physicists seek a quantum field theory of the gravitational force—a theory in which the smallest bundle of a